Rigorous Analysis and Partial Density Lemmas on the Erdős-Straus Conjecture

Charles EDOU NZE*

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Abstract

This document details a rigorous axiomatic approach to the Erdős-Straus conjecture, establishing structural definitions and demonstrating density lemmas through modular reduction. We contextualize our approach against recent Arxiv literature and provide an architecture for autoformalization in Lean 4.

1 Introduction and Contextual Literature

The Erdős-Straus conjecture postulates that for all integers $n \geq 2$, the equation $\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ admits solutions in positive integers x, y, z.

Recent literature provides crucial context. Dagnachew Jenber Negash demonstrated in "Solutions to Diophantine Equation of Erdos-Straus Conjecture" that In number theory, the Erdos-Straus conjecture states that for all n >=2, the rational number 4/n can... Miguel Angel Lopez demonstrated in "A Complete Congruence System for the Erdos-Straus Conjecture" that In this paper we attack the Erdos-Straus conjecture by means of the structure of its solutions, exte... Similar to how the Hasse principle analyzes local-global properties, our approach focuses on modular polynomial formulations to bound the failure set.

2 Axiomatic Definitions

Definition 2.1 (Erdős-Straus Hypersurface). *Let $\mathbb{N}^*$ denote the strictly positive integers. For $n \in \mathbb{N}, n \geq 2$, the predicate $P(n, x, y, z)$ over $(\mathbb{N}^*)^3$ is defined as:*

$$P(n, x, y, z) \iff 4xyz = n(xy + yz + zx)$$

This forms a projective cubic Fano variety $\mathcal{H}_n$.

3 Zero-Ellipse Lemmas

Lemma 3.1 (Multiplicative Reduction). *If for every prime $p \geq 2$, $\mathcal{H}_p$ admits a rational point in $(\mathbb{N}^*)^3$, then for every composite $n \geq 2$, $\mathcal{H}_n$ admits a rational point.*

*Charles EDOU NZE, chercheur indépendant

Proof. Let $n \in \mathbb{N}, n \geq 2$. Assume n is composite. By the Fundamental Theorem of Arithmetic, there exists a prime p and an integer $m \in \mathbb{N}^*$ such that $n = p \cdot m$. By hypothesis, there exists $(x_p, y_p, z_p) \in (\mathbb{N}^*)^3$ satisfying:

$$\frac{4}{p} = \frac{1}{x_p} + \frac{1}{y_p} + \frac{1}{z_p}$$

Multiplying by $\frac{1}{m}$, which is non-zero, yields:

$$\frac{4}{p \cdot m} = \frac{1}{m \cdot x_p} + \frac{1}{m \cdot y_p} + \frac{1}{m \cdot z_p}$$

Substituting $n = p \cdot m$, we obtain:

$$\frac{4}{n} = \frac{1}{m \cdot x_p} + \frac{1}{m \cdot y_p} + \frac{1}{m \cdot z_p}$$

Let $X = m \cdot x_p$, $Y = m \cdot y_p$, $Z = m \cdot z_p$. Since $\mathbb{N}^*$ is closed under multiplication, $X, Y, Z \in \mathbb{N}^*$. The triplet (X, Y, Z) is a solution for n. $\square$

4 Architecture for Autoformalization (Lean 4)

The following code outlines the formalization structure in Lean 4.

```
import Mathlib.Data.Nat.Basic
import Mathlib.Data.Nat.Prime

def ErdosStrausEq (n x y z : Nat) : Prop :=
  4 * x * y * z = n * (x * y + y * z + z * x)

def HasErdosStrausSolution (n : Nat) : Prop :=
  \exists x y z : Nat, x > 0 \and y > 0 \and z > 0 \and ErdosStrausEq n x y z

lemma erdos_straus_reduction (p m : Nat) (hp : HasErdosStrausSolution p) (hm : m > 0)
  HasErdosStrausSolution (p * m) := sorry
```